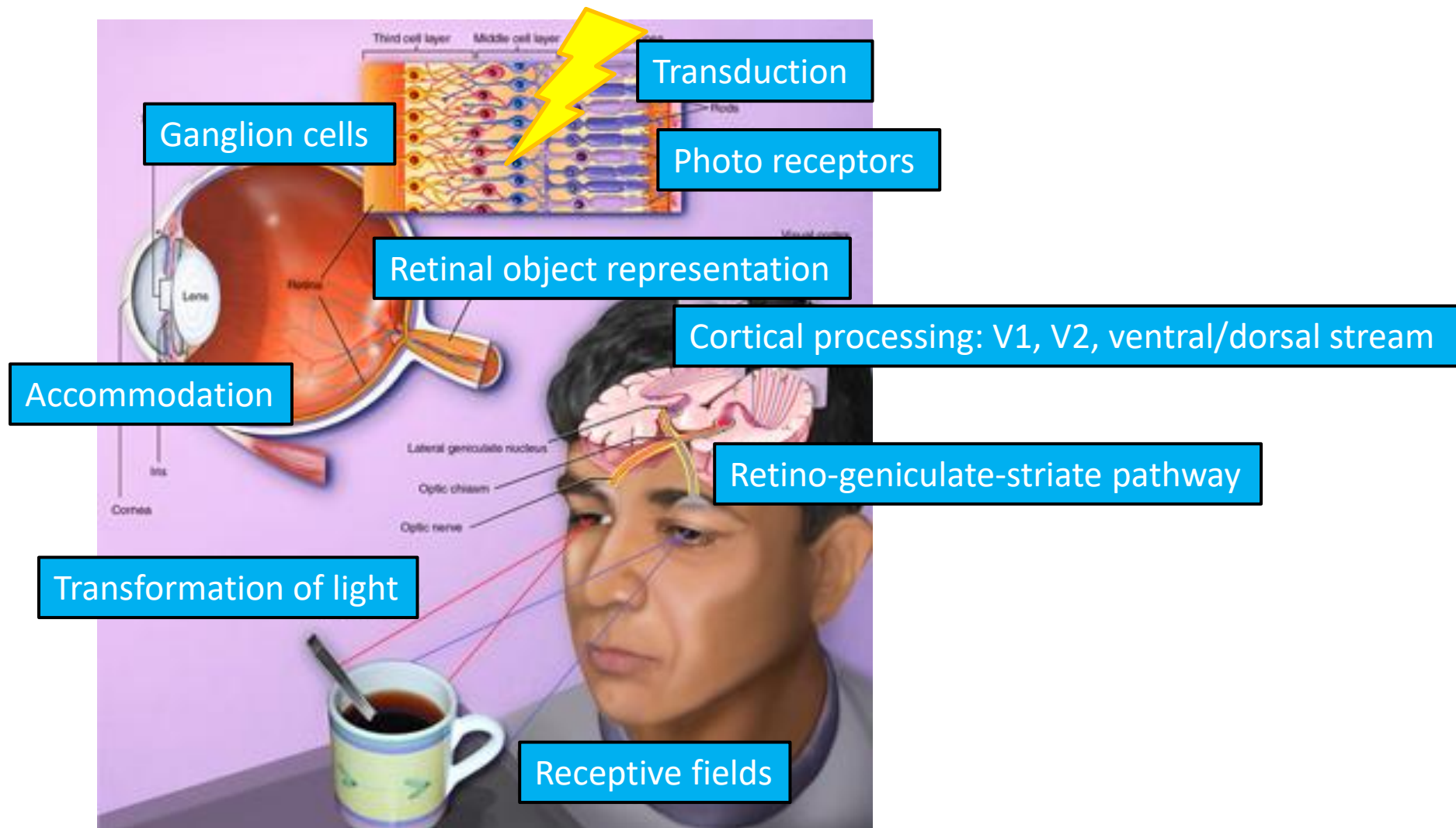


The psychology of perception

Prof Anna Grubert

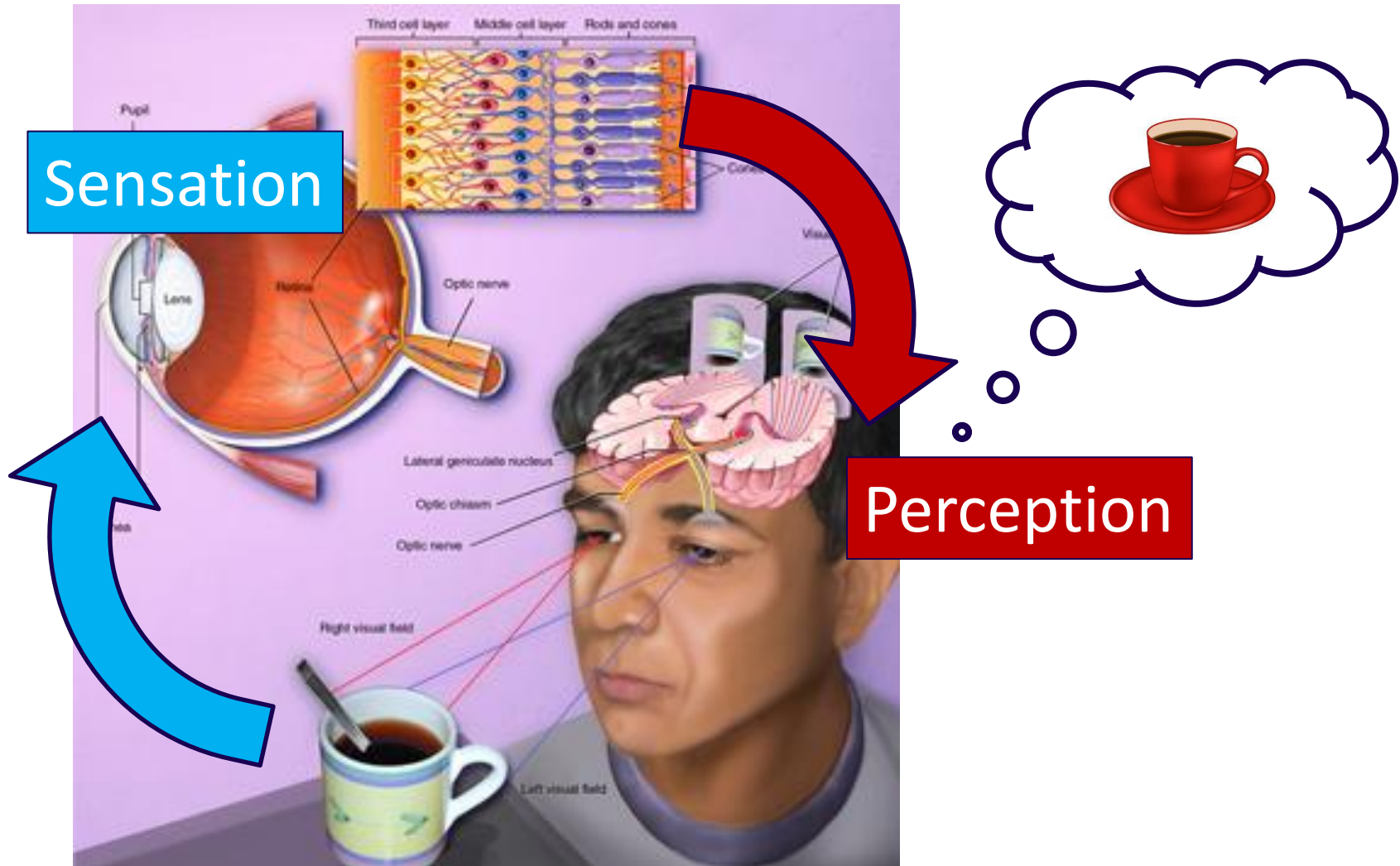
anna.k.grubert@durham.ac.uk

Last week: The physiology of visual processing



Today:

- How do retinal images and electrical signals become a coffee mug?



Sensation and perception

- How do retinal images and electrical signals become a coffee mug?



Sensation

The **un-interpreted** sensory impressions created by the detection of a stimulus.



Perception

The (psychological, cognitive) processes of **making sense** of the sensations.

Sensory elements

- Structuralism (late 19th century)
 - The study of the **elements of consciousness**.
 - **Conscious experience (perception)** can be broken down into **basic elements (sensory elements)**, which can then be combined again to describe any human experience.
 - Father of Structuralism:
Wilhelm Wundt, 1879,
started the first psychological
laboratory in Leipzig, Germany.



Sensory elements

- Structuralism
 - Method: **Introspection**



Introspection report from **Edward Titchener**, 1901:

“Thus the ‘**taste**’ of lemonade is made up of a **sweet** taste, an **acid** taste, a **scent** (the fragrance of lemon), a sensation of **temperature** and a **pricking** (cutaneous) sensation. The ‘**taste**’ of limewater is made up of a weakly **sweet** taste, a sensation of **nausea** (organic sensation), a sensation of **temperature** and a **biting** (cutaneous) sensation.”

*Cataloguing **sensation** and **perception***

Sensory elements

- Structuralism – summary

- Goal

Describe elements of perception.



- **Perception** = Sum (combination) of **sensory elements**

Coffee mug = **Round** and **cupped** appearance with a **handle**; **roasted** and **bitter** smell; **hot** touch; ...

- Evaluation

(+) First “**school of thought**” in psychology

(-) **Validity** (no measures, purely descriptive)

(-) **Reliability** (observations are not consistent and constant)

(-) **Objectivity** (observations depend on observer; biased)

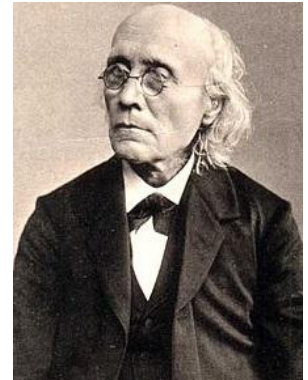
Measuring the elements of perception

- Psychophysics (1/3)

- **Gustav Fechner, 1860**

Book: “Elements of Psychophysics”

Measuring rather than describing the elements of perception.



- Basic idea

Measuring the absolute (perception) threshold.

(Just noticeable intensity threshold to detect a stimulus.)

1) Method of adjustment

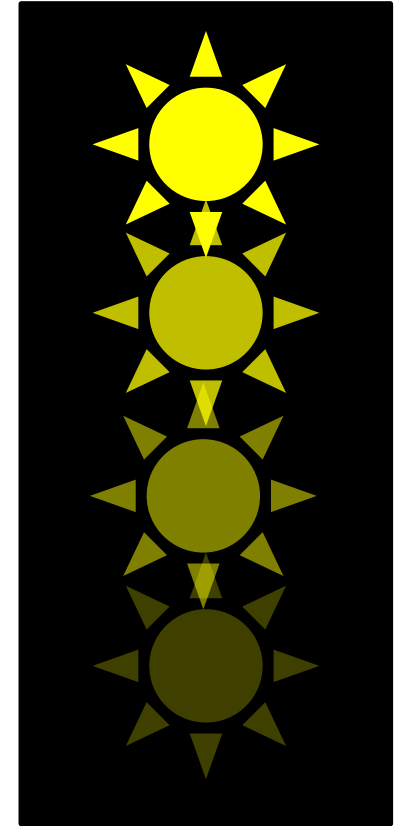
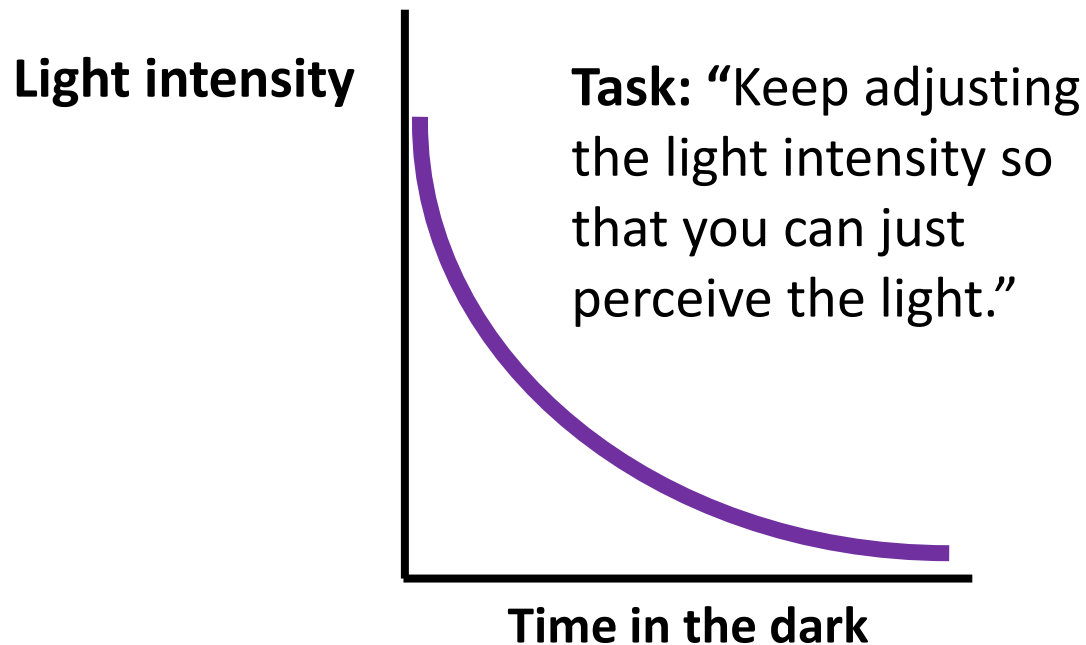
2) Method of limits

3) Method of constant stimuli

Measuring the elements of perception

1) Method of adjustment

Participants adjust the intensity of a test light, until they are **just able to perceive** the light.



Measuring the elements of perception

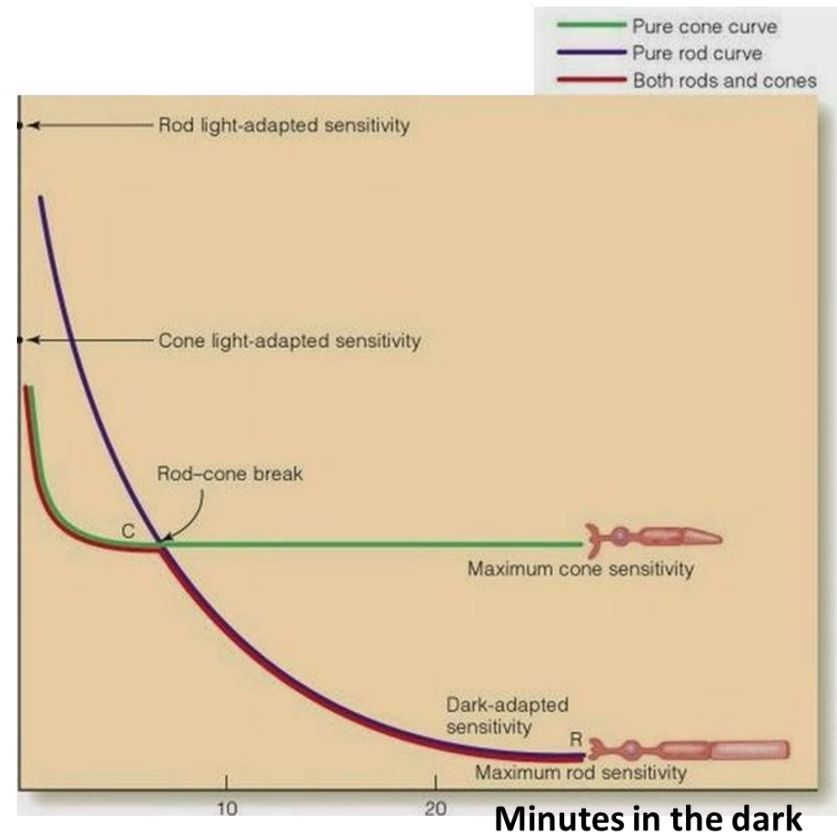
Dark adaptation experiment

Adjustment of light intensity over time in darkness.

Note: Just an example for the method of adjustment (not actually conducted by Fechner)

Sensitivity to light
Low (day)

High (night)



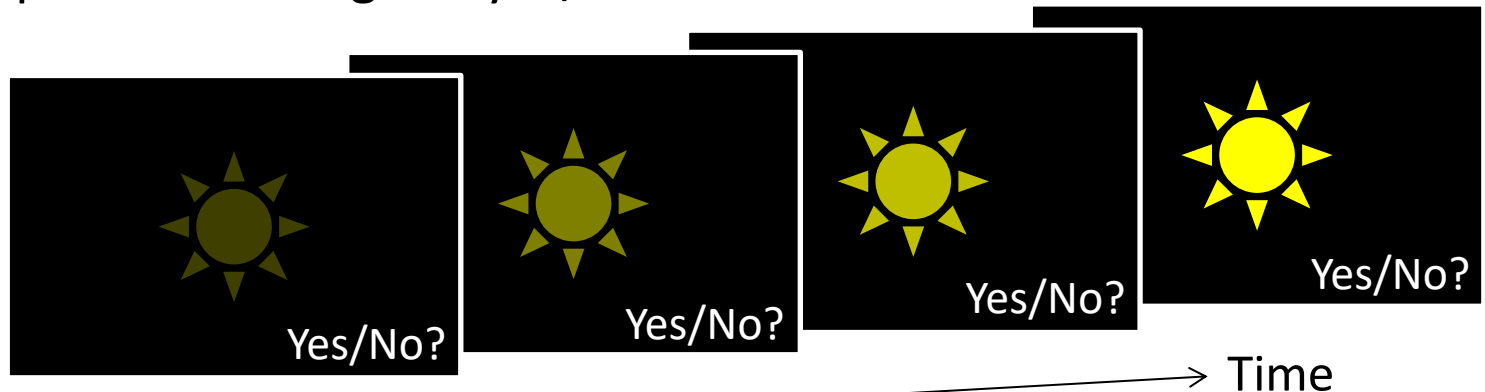
Measuring the elements of perception

2) Method of limits

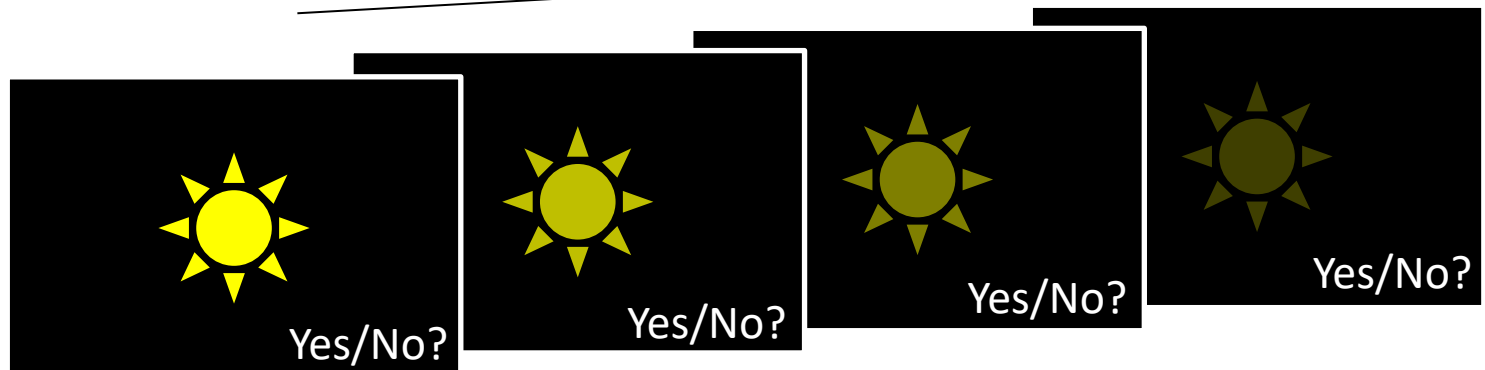
Participants are presented with trials of increasing/decreasing light intensity.

Task: Can you perceive the light – yes/no?

Trial with
ascending
light
intensity



Trial with
descending
light
intensity



Measuring the elements of perception

— Absolute perception
threshold in each trial

A = Ascending light intensity

D = Descending light intensity

Y/N = Stimulus perceived yes/no

Stimulus Intensity	A	D	A	D	A	D	A	D
9		Y						Y
8		Y				Y		Y
7		Y		Y		Y		Y
6		Y		Y		Y		Y
5		Y		Y		Y		Y
4	Y	Y		Y		Y		Y
3	N	Y	Y	Y	Y	Y	Y	Y
2	N	N	N	Y	N	N	N	Y
1	N	N	N	N	N	N	N	N
0	N	N	N	N	N	N	N	N
	3.5	2.5	2.5	1.5	2.5	2.5	1.5	1.5

Mind the variance across trials (multiple trials are averaged to obtain mean threshold).

Measuring the elements of perception

— Absolute perception threshold in each trial

A = Ascending light intensity

D = Descending light intensity

Y/N = Stimulus perceived yes/no

Stimulus Intensity	A	D	A	D	A	D	A	D
9		Y						
8		Y				Y		Y
7		Y		Y		Y		Y
6		Y		Y		Y		Y
5		Y		Y		Y		Y
4	Y	Y		Y		Y		Y
3	N	Y	Y	Y	Y	Y	Y	Y
2	N	N	N	Y	N	N	N	Y
1	N	N	N	N	N	N	N	N
0	N		N		N		N	
	3.5	2.5	2.5	1.5	2.5	2.5	1.5	1.5

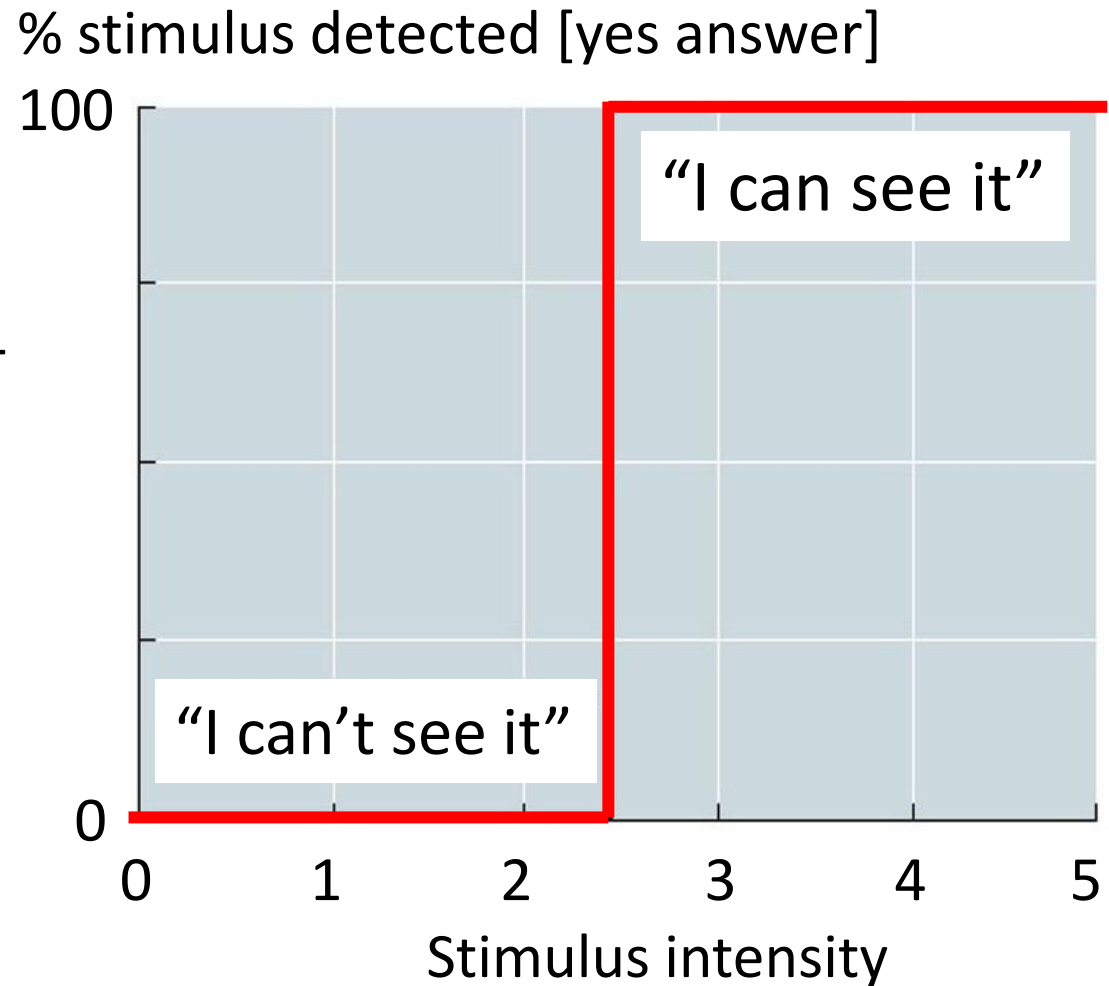
Mean Threshold = 2.25 (sum of thresholds in each trial [3.5+2.5+1.5+2.5+2.5+1.5+1.5] / number of trials [8])

Mind the variance across trials (multiple trials are averaged to obtain mean threshold).

Measuring the elements of perception

Result:

Relatively crude threshold – stimulus is either detected (100%) or not (0%).

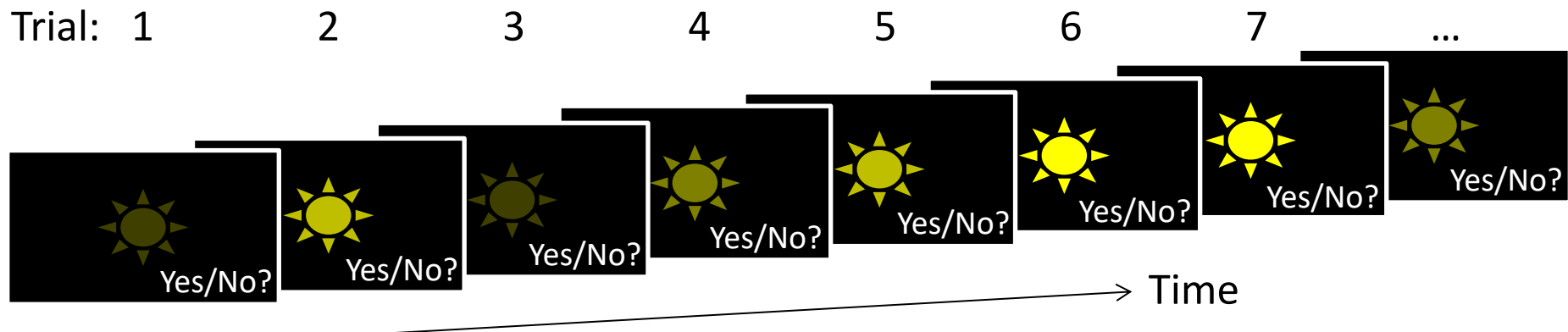


Measuring the elements of perception

3) Method of constant stimuli

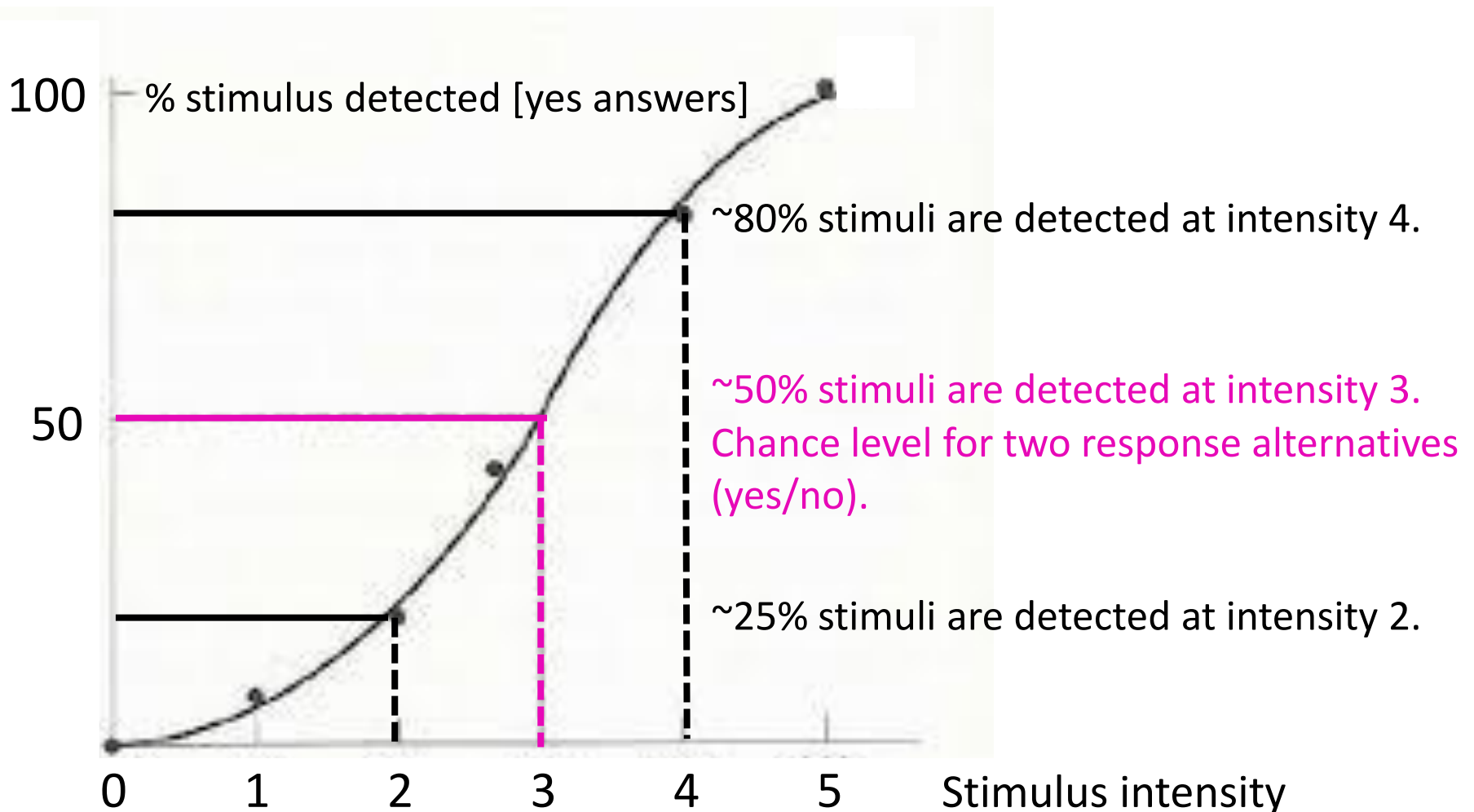
Like the method of limits (same task), but:

- many more trials per light intensity.
- randomised light intensity across trials, i.e., each light input is a separate trial (no confound of stimulus order).



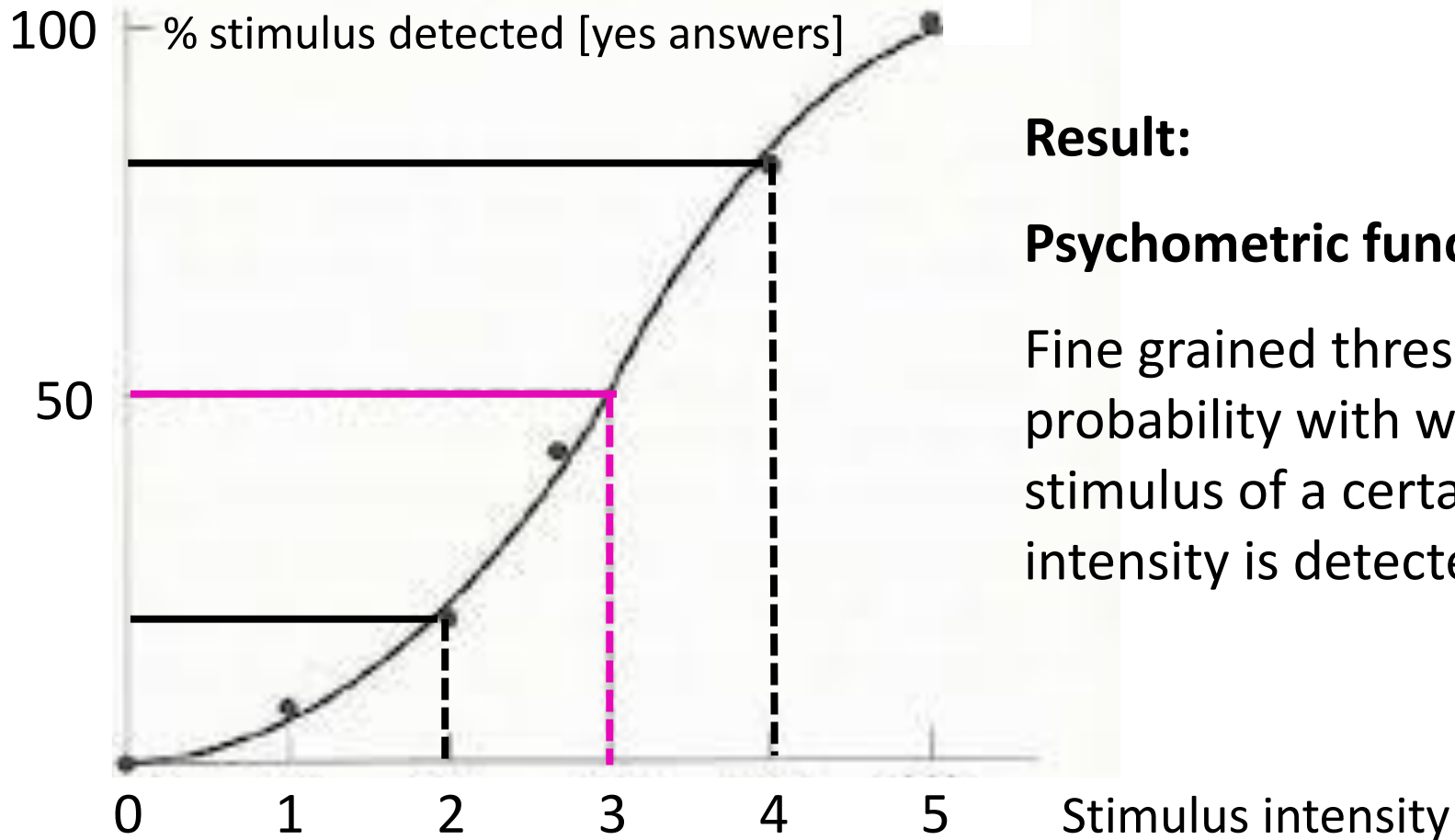
Measuring the elements of perception

Percentage of **yes answers** is plotted as a function of **stimulus intensity**.



Measuring the elements of perception

Percentage of **yes answers** is plotted as a function of **stimulus intensity**.



Result:

Psychometric function.

Fine grained threshold - probability with which a stimulus of a certain intensity is detected.

Measuring the elements of perception

- Psychophysics – interim summary

Gustav Fechner (1860)

- How much does it take to detect a stimulus?
Measurement of the absolute perception threshold.
Methods of adjustment, limits, and constant stimuli.
- How much does it take to tell two stimuli apart?

Measuring the elements of perception

- Psychophysics (2/3)



- **Ernst Weber**, 1834

- Basic idea

Measuring the difference threshold.

(Minimum intensity difference to discriminate two stimuli.)

- **Weber's law**

The change in a stimulus, to discriminate it from another stimulus, is a constant ratio of that original stimulus.

Measuring the elements of perception

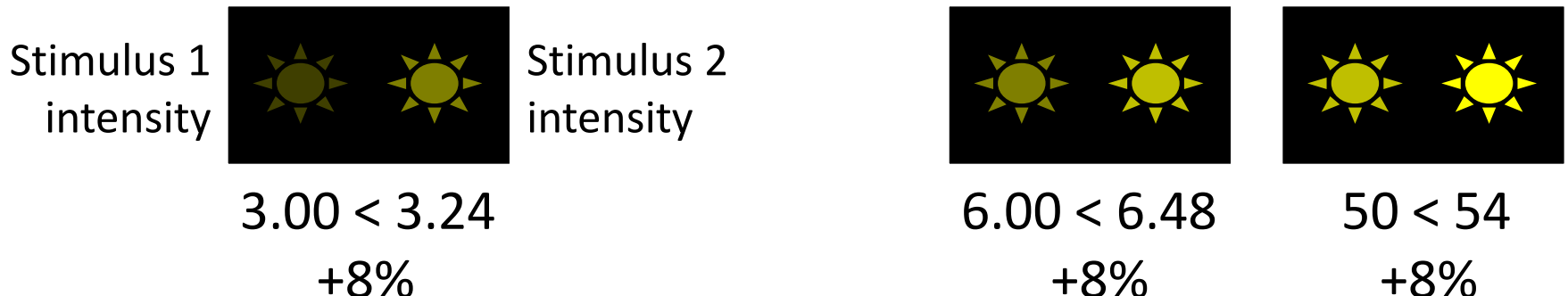
- Psychophysics (2/3)

- **Weber's law**

i.e., the brighter light 1 is, the greater is the required brightness change for light 2 to be perceived as different.

... required brightness change has a constant ratio.

Difference threshold for light intensity = 0.08 (8%)



Measuring the elements of perception

- **Difference threshold:**

- The change in a stimulus, to discriminate it from another stimulus, is a constant ratio of that original stimulus

When you are in a pub with many other people you must shout to be heard, while a whisper works when you are in the library.



Measuring the elements of perception

- Psychophysics (2/3)

- Weber's law in the real world

Can you tell the difference between:

- coffee with 1 or 9 cubes of sugar?
- coffee with 8 or 9 cubes of sugar?



Measuring the elements of perception

- Psychophysics – interim summary

Gustav Fechner (1860)

- How much does it take to detect a stimulus?

Measurement of the absolute perception threshold.

Methods of adjustment, limits, and constant stimuli.

Ernst Weber (1834)

- How much does it take to tell two stimuli apart?

Weber's law: This depends (in a fixed ratio) on the strength of the original stimulus.

- How does a stimulus change feel subjectively?

Measuring the elements of perception

- Psychophysics (3/3)

- **Stanley Stevens, 1957**



- Basic idea

Subjective magnitude estimation.

(Measuring the relation between stimulus intensity [objective measure] and perceived intensity [subjective experience]).

An increase in the **perceived** stimulus intensity can be

larger or smaller

response expansion

response compression

than the increase in the **measured** stimulus intensity.

Measuring the elements of perception

- Psychophysics (3/3)

Steven's power function:

$$P = KS^n$$

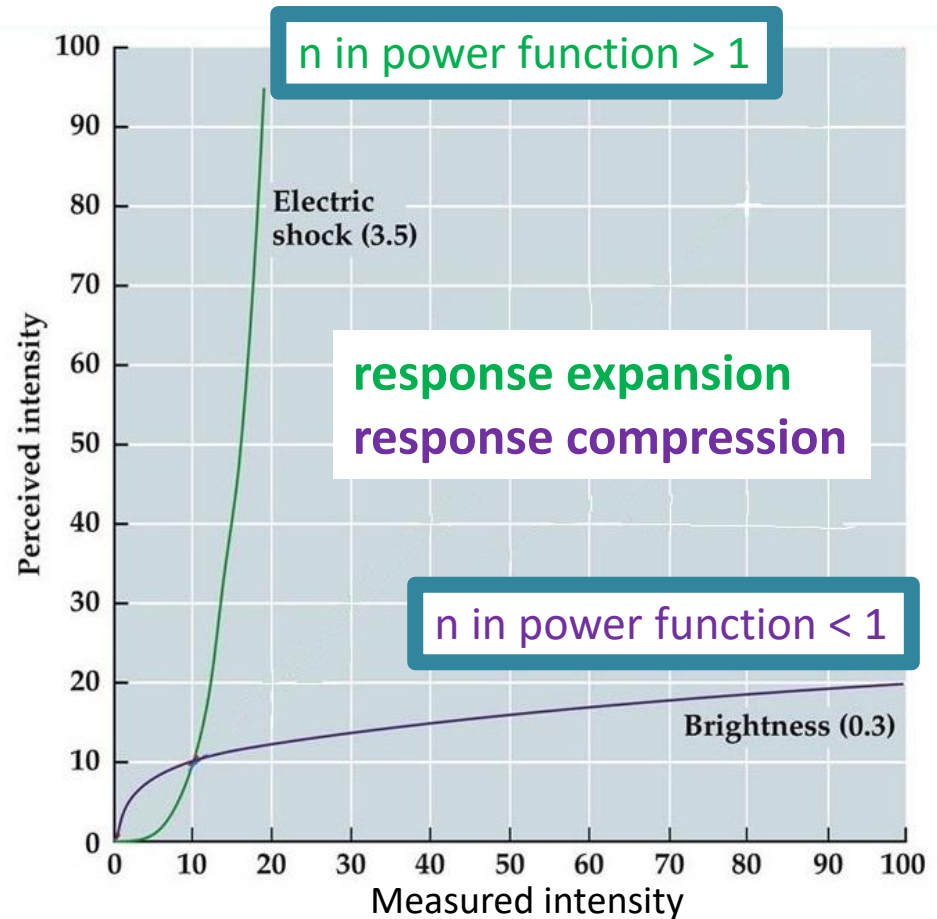
With

P = perceived intensity

K = constant (scales values to size)

S = stimulus intensity

n = exponent



An increase in the **perceived** stimulus intensity can be **larger** or **smaller** than the increase in the **measured** stimulus intensity.

Measuring the elements of perception

- Psychophysics (3/3)

Steven's power function:

$$P = KS^n$$

With

P = perceived intensity

K = constant (scales values to size)

S = stimulus intensity

n = exponent

(given)

$$K = 5$$

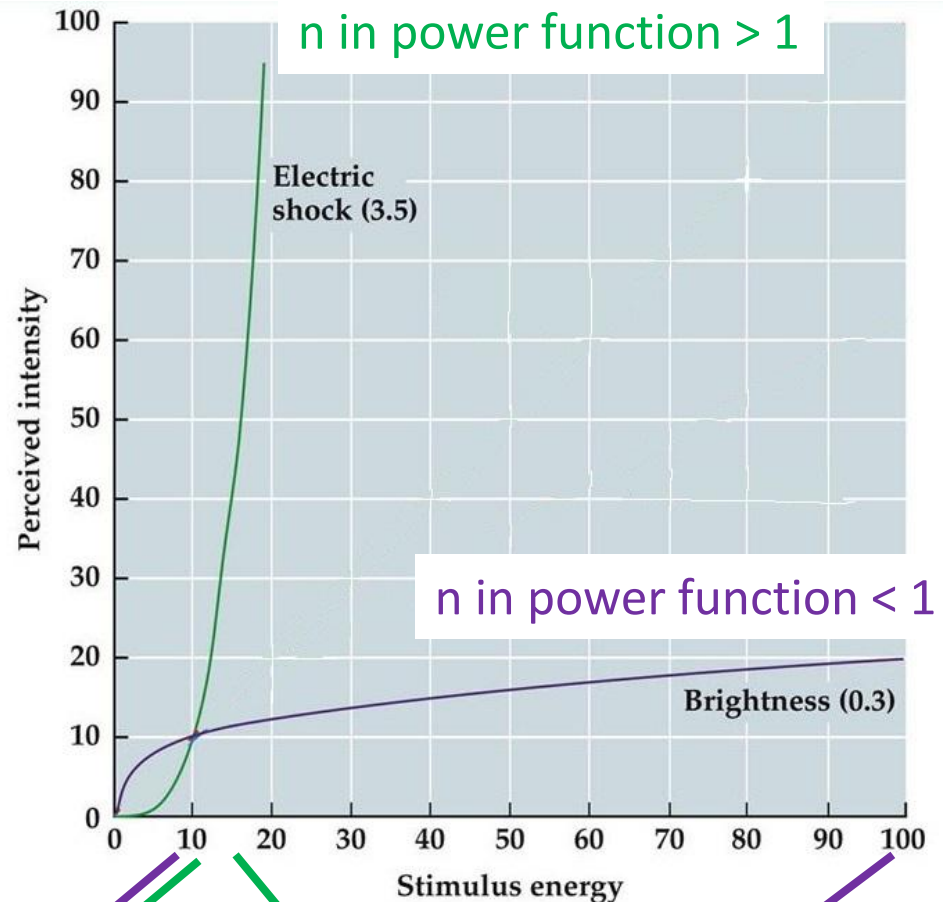
$$K = 0.0031$$

$$P = 5 * 10^{0.3} = 10$$

$$P = 0.0031 * 10^{3.5} = 10$$

$$P = 5 * 100^{0.3} = 19.9$$

$$P = 0.0031 * 15^{3.5} = 40.5$$



Measuring the elements of perception

- Psychophysics – interim summary

Gustav Fechner (1860)

- How much does it take to detect a stimulus?

Measurement of the absolute perception threshold.

Methods of adjustment, limits, and constant stimuli.

Ernst Weber (1834)

- How much does it take to tell two stimuli apart?

Weber's law: This depends (in a fixed ratio) on the strength of the original stimulus.

Stanley Stevens (1957)

- How does a stimulus change feel subjectively?

More, or less, or exactly as strong, as the measurable physical change.



n in power function = 1

Measuring the elements of perception

- Psychophysics – summary

- Goal

Objective **measurement** of perceptual elements.



- **Perception** = Sum of **measurable sensations**.

Coffee mug = A number of measurable intensities (**size**, **luminance**, **temperature**, ...).

- Evaluation

(+) Valid, reliable, and objective measures (=hard data).

(-) **No theoretical account** of perception (i.e., unclear how we add meaning to our sensations).

Theoretical accounts of perception (1*)

- **James Gibson, 1966**
 - **Ecological theory** of perception
 - Rationale



Perception is direct (perception = sensation).

“The real world is rich in sensory information and provides sufficient context for the visual systems to directly perceive what is there.”

Perception must be investigated in a **natural environment**.

* More theoretical accounts of perception will follow in Lecture 4.

Theoretical accounts of perception (1)

- Basic idea

Perception...

...takes place in the **optic array** (light in the environment).

... is **directly** based on **invariant information** in the visual field, which is extracted by the **observer's movement**.

i.e., As an observer moves, the optic array becomes **ambient** (the light surrounding the observer changes), but some information remains constant (**invariant**).

Theoretical accounts of perception (1)

- Examples of invariant visual information (Gibson, 1966)
 - Optic flow pattern

The **focus point** of a driver remains **motionless** while the rest of the visual field appears to move away from this point.

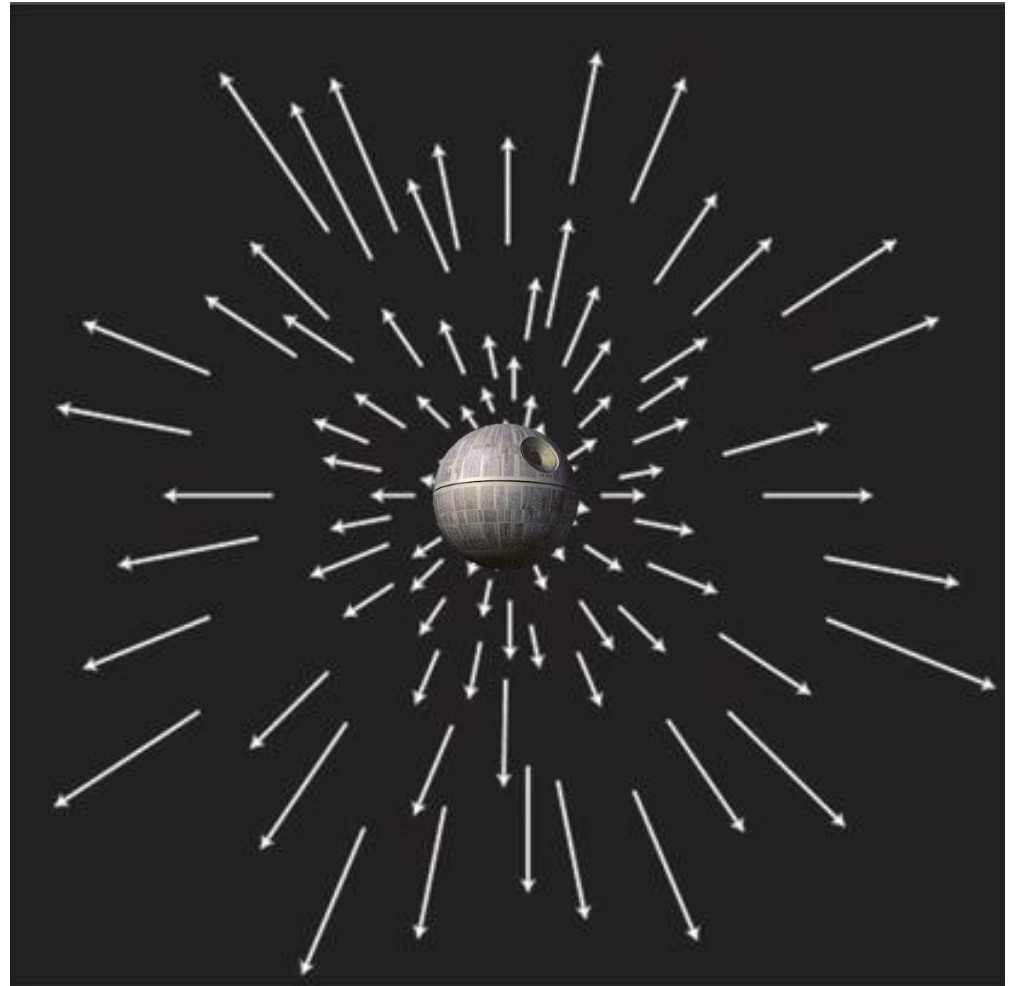


Theoretical accounts of perception (1)

- Examples of invariant visual information (Gibson, 1966)

- Optic flow pattern

The **focus point** of a pilot remains **motionless** while the rest of the visual field appears to move away from this point.



Theoretical accounts of perception (1)

- Examples of invariant visual information (Gibson, 1966)
 - Texture gradient

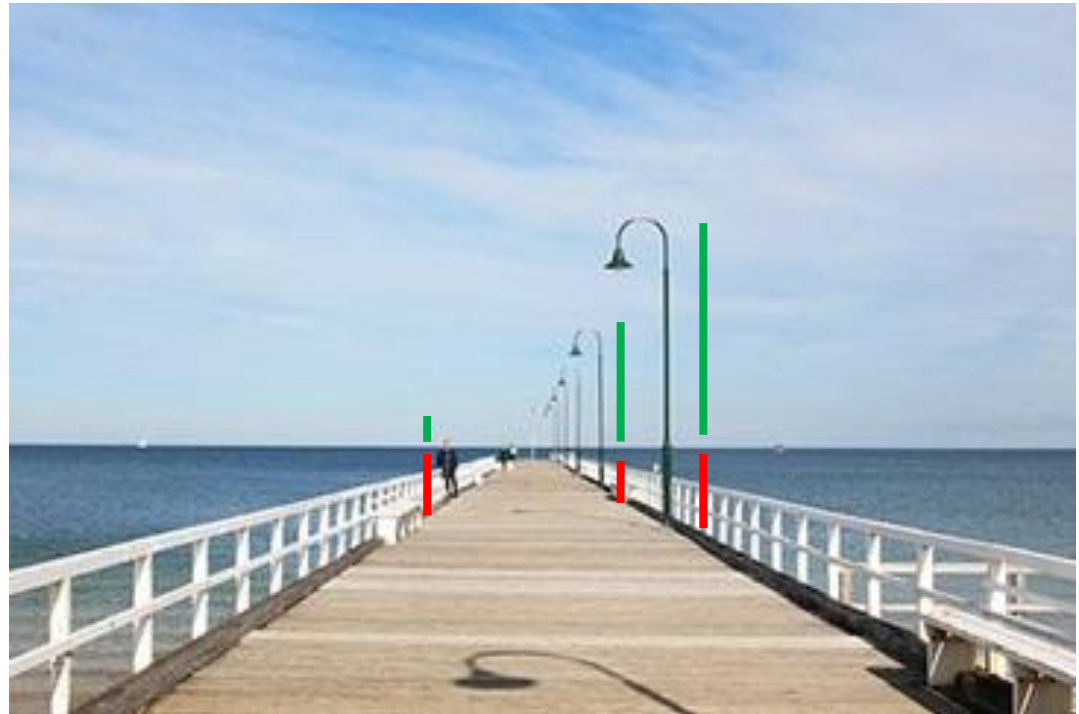
The elements of a **textured ground** are **denser in the distance**.



Theoretical accounts of perception (1)

- Examples of invariant visual information (Gibson, 1966)

- Horizon ratio



The **proportion of the object** **above** and **below** the horizon is **constant** for objects of the same size standing on the same ground.

Theoretical accounts of perception (1)

- Examples of invariant visual information (Gibson, 1966)

- ...and many other invariants:

Gravity



Straight lines



Reflectance of different areas



Theoretical accounts of perception (1)

- Ecological theory of perception – summary
 - Goal
Explain how we attach **meaning** to **sensory input**.
 - **Perception** = Directly offered by the world (**sensation**).

Affordance of objects: “When we combine our physical and psychological states with our constantly changing optic arrays, we are enabled to recognise not only **what the object** is but **what it does**.”



Coffee mug = Affords (offers) grabbing it (**handle**), drinking from it (**cupped shape**), enjoying it (**smell**), ...

Theoretical accounts of perception (1)

- Ecological theory of perception – summary

- Evaluation

(+) **Theoretical account** (tries to explain how perception and object recognition work).

(+) Assumes that perception, offered by the real world, is...

...**fast**, readily available, and effortless.

...always **accurate**.

Theoretical accounts of perception (1)

- Ecological theory of perception – summary

- Evaluation

(+) **Theoretical account** (tries to explain how perception and object recognition work).

(+) Assumes that perception of the real world, is...
...**fast**, readily available.
...always effortless.
...always accurate.

(-) Reality check next week!

Lecture 3 - Evaluation of today's accounts

- Today's accounts

Structuralism (Wundt, Titchener)

Psychophysics (Fechner, Weber, Stevens)

Ecological theory of perception (Gibson)

... assume that perception is automatic.

Perception equals **sensation**.

- They are

- **Bottom-up accounts.**

Stimulus-guided perception (information flow from the stimulus to the brain),

i.e., information gathered by the sensory receptors determines perception completely.

Summary

- We have looked at **3 bottom-up accounts** of perception
 - 1) **Structuralism** (How to describe sensation).

Introspection
 - 2) **Psychophysics** (How to measure sensation).

Absolute perception threshold (Fechner's methods of adjustment, limits, and constant stimuli).
Difference threshold (Weber's law).
Magnitude estimation (Steven's power function).
 - 3) Gibson's **ecological theory** of perception (How to make sense of sensation).

Direct perception in the optic array is based on invariant visual information.
Perception is based on object affordances.

Test yourself – learning goals

- After reviewing this lecture, you should be able to
 - Explain what the difference between sensation and perception is.
 - Debate how Structuralists versus Psychophysicists studied and interpreted perception.
 - Name 3 Psychophysicists and explain their measures.
 - Compare the methods of adjustment, limits and constant stimuli.
 - Explain what Gibson meant by direct perception and object affordance.
 - Debate why today's accounts are bottom-up accounts of visual perception.

Test yourself – multiple choice

1. Which of the following key words is not related to psychophysics?

- a) Method of adjustment.
- b) Weber's law.
- c) Affordance.
- d) Magnitude estimation.

Test yourself – multiple choice

2. Which of the following scenarios is not in line with Weber's law?

- a) When you have a messy room, it is not sufficient to pick up one or two items, but when you pick up 20, the room will appear less messy.
- b) When you enter a house party and want to check whether your mate is already there you shout their name as loud as you can.
- c) In the cinema, turning down the brightness of your mobile phone screen to the dimmest level will be sufficient.
- d) All of them are.

Test yourself – multiple choice

3. Which of the following statements does not match the concept of direct perception (Gibson, 1966)?

- a) Perception is effortless.
- b) Perception is error-prone.
- c) Perception is free of interpretation.
- d) Perception is based on invariant information.

See you on Friday...



Texture gradient - London tube style!